VisioSphere — Strong Review of Related Literature

Peer-reviewed sources only · 2020 or later · each source opened and verified on the publisher page · each reports quantitative results. Preprints (arXiv / bioRxiv), theses, conference abstracts and papers without reported metrics were excluded. 32 sources. Compiled 9 September 2026.

Inclusion bar applied

A source was accepted only if **all four** held: (1) peer-reviewed journal or major conference proceedings; (2) published 2020 or later; (3) topically direct to a measurement VisioSphere actually performs, or to the generalisation claim; (4) reports real numbers — accuracy, AUC, sensitivity, false-alarm rate, or measured threshold values. Nine otherwise-relevant candidates were rejected under this bar; they are listed in §5 so you know they were considered and why they were dropped.

1. Lying / fall geometry — the LYING_EVIDENCE gate

These back the four conditions of the new positive-evidence gate: lower-body visibility, the torso-normalised vertical stack ratio, the bounding-box aspect limits, and the hip-extension / torso-angle alternatives.

Scale-invariant body-segment ratios (backs your stack ratio)

[1] K. Han, Q. Yang, and Z. Huang, “A Two-Stage Fall Recognition Algorithm Based on Human Posture Features,” *Sensors*, vol. 20, no. 23, art. 6966, 2020.
https://doi.org/10.3390/s20236966

Supports: The closest published analogue to your vertical stack ratio. Computes a **spine ratio** = spine vector length ÷ waist-width vector length from OpenPose 2D keypoints — the same construction as yours: divide one body measurement by another measured in the same frame, so camera distance cancels. Combined with segment deflection angles to vertical.

Quotable figures: Decision boundary **RB = 1.0**; ADL frames cluster above **1.2**, fall frames below **1.2**. Angle thresholds 30° / 50°. Accuracy **97.34%**, precision 98.50%, recall 97.33%, F1 97.91%.

[2] B. M. V. Guerra, S. Ramat, G. Beltrami, and M. Schmid, “Automatic Pose Recognition for Monitoring Dangerous Situations in Ambient-Assisted Living,” *Frontiers in Bioengineering and Biotechnology*, vol. 8, art. 415, 2020.
https://doi.org/10.3389/fbioe.2020.00415

Supports: The clearest published *statement* of why you normalise. Four-class standing / sitting / lying / dangerous-sitting classification using joint angles plus vertical joint coordinates **explicitly normalised by subject height** “to eliminate subject-specific variability.” Also documents your occlusion problem: sitting frames are noisy precisely because joints are hidden by other body parts.

Quotable figures: 602,530 frames, 12 subjects, held-out-subject test. Mean accuracy **83.9% ± 0.73%**; **lying 94.3%** (AUC 98.5), standing 93.8%, sitting 86.9%. Sitting↔dangerous-sitting confusion 6.01% / 2.91%.

Joint angles for lying vs sitting vs standing

[3] V. Kong, S. Soeng, M. Thon, W.-S. Cho, A. Nayyar, and T.-K. Kim, “PIFR: A novel approach for analyzing pose angle-based human activity to automate fall detection in videos,” *PLOS ONE*, vol. 20, no. 6, art. e0325253, 2025.
https://doi.org/10.1371/journal.pone.0325253

Supports: Nine pose-angle features drive standing-vs-lying classification, including **torso angle to vertical, hip angle to horizontal and left/right knee-joint leg angles** — the same angle family your gate uses. Authors state the features are built to “normalize the input data, irrespective of variations in object bounding box sizes and locations.”

Quotable figures: Precision **88.8%**, recall **94.1%**, F1 91.4%, specificity 95.6%, **false-positive rate 4.4%**, 23 FPS. URFD + Multiple Cameras Fall Dataset.

[4] H. Shi, X. Wang, and J. Shi, “Fall Detection Algorithm Using Enhanced HRNet Combined with YOLO,” *Sensors*, vol. 25, no. 13, art. 4128, 2025.
https://doi.org/10.3390/s25134128

Supports: Bounding-box aspect ratio limits **combined with** a torso-tilt-to-ground threshold, from COCO-17 keypoints — structurally your LYING_ASPECT_MAX plus the torso-angle > 65° branch. Publishes actual cut points, which most papers do not.

Quotable figures: A = $(y_{max}–y_{min})/(x_{max}–x_{min})$; walking **1.5** ≤ **A < 3**; fall requires **A < 1** AND descent **> 350 px/s** AND **trunk tilt > 75°**. Accuracy **> 95%** at 18.1 fps.

Keypoint-confidence gating and false-alarm reduction

[5] H. Rehouma and M. Boukadoum, “Fall Detection by Deep Learning-Based Bimodal Movement and Pose Sensing with Late Fusion,” *Sensors*, vol. 25, no. 19, art. 6035, 2025.
https://doi.org/10.3390/s25196035

Supports: Your single best citation for **MANDATORY 1** (lower-body visibility at conf ≥ 0.45) and for temporal confirmation. Excludes landmarks below a visibility threshold *before* any geometric reasoning, and requires a run of consecutive frames before alarming — the same architecture as LYING_EVIDENCE_MIN_FRAC / MIN_FRAMES.

Quotable figures: Visibility cut **0.50**; aspect < 1 upright, > 1 prone; **30 consecutive frames (1 s)** to confirm. F1 **97.3%**, FPR **3.6%** vs **8.9%** vision-only and **11.3%** IMU-only. False positives from “abrupt sitting” explicitly suppressed.

[6] L. Qi and H. Sun, “Multistage fall detection framework via 3D pose sequences and TCN integration,” *Scientific Reports*, vol. 15, art. 27832, 2025.
https://doi.org/10.1038/s41598-025-11325-y

Supports: Evidence that your false alert of 09-09 (resident bent over a corner table) is a **recognised, quantified field-wide problem**, not a defect unique to VisioSphere. The paper deliberately stress-tests against sitting down, squatting and shoe-removal.

Quotable figures: NTU RGB+D: **890 fall samples vs 2,800 deliberately similar non-fall samples**. Accuracy 99.87%, macro-F1 97.68%, **FP and FN rates as low as 0.13%**. Best with 16 joints.

[7] A. Bustamante, L. M. Belmonte, R. Morales, A. Pereira, and A. Fernández-Caballero, “Bridging the Appearance Domain Gap in Elderly Posture Recognition with YOLOv9,” *Applied Sciences*, vol. 14, no. 21, art. 9695, 2024.
https://doi.org/10.3390/app14219695

Supports: Problem framing and baseline: standing / sitting / lying discrimination in real elderly subjects from a single camera is a well-posed, high-accuracy task. **Note honestly:** bounding-box only — no keypoints, angles or ratios. It supports the task, not your gates.

Quotable figures: Best config accuracy **98.41%**, F1 0.97; lying precision 0.93 / recall 0.95. Useful warning: the virtual-to-real transfer collapses to **70.64% accuracy, F1 0.25**.

2. Agitation — reversal counting and the CV rhythm test

Extrema / reversal detection in a pose trajectory

[8] H. L. Cornman, J. Stenum, and R. T. Roemmich, “Video-based quantification of human movement frequency using pose estimation: A pilot study,” *PLOS ONE*, vol. 16, no. 12, art. e0261450, 2021.
https://doi.org/10.1371/journal.pone.0261450

Supports: Validates your **reversal-counting front end** directly: detecting local minima and maxima in a pose-estimated hand/limb trajectory is an accurate way to identify discrete movement events and derive movement frequency.

Quotable figures: Event-time correlation with manual annotation **r > 0.99** across all tasks at 1–4 Hz; mean frequency error ≤ **0.03 Hz**; **993 vs 1002** finger taps detected (pose vs manual); typical event-time error 1–2 frames.

[9] F. L. Marshall, S. Zhang, and B. Scotney, “Automatic Assessment of the Type and Intensity of Agitated Hand Movements,” *Journal of Healthcare Informatics Research*, vol. 6, no. 4, pp. 401–422, 2022.
https://doi.org/10.1007/s41666-022-00120-3

Supports: The closest published system to your agitation module: agitated hand movement in a dementia context, quantified from RGB video by tracking a hand centroid as a 1-D trajectory, **peak detection at local extrema**, intensity expressed as **repetitions per minute**. Backs both your reversal counting and your rate gate (AGITATION_OSC_MIN_RATE).

Quotable figures: Metronome-paced intensities **30, 50, 70, 90 rpm**; 20 participants, 400 × 30 s sequences; five movement types. Type accuracy 80.0%, **intensity accuracy 92.5%**, both correct 75.25%. Type window 6 s; **intensity window 12 s with 5 s overlap**.

Coefficient of variation as a rhythmicity index

[10] A. Numata, Y. Terao, K. Sugawara, Y. Ugawa, and T. Furubayashi, “Differences in the movement phase condition and sensory inputs on temporal synchronization and continuation during bilateral foot-tapping tasks,” *Frontiers in Human Neuroscience*, vol. 19, art. 1518230, 2025.
https://doi.org/10.3389/fnhum.2025.1518230

Supports: Your methodological precedent: **CV of inter-tap intervals used as the primary index of rhythmic timing stability** in repetitive human movement. This is the same statistic you apply to inter-reversal intervals.

Quotable figures: CV of ITI, SC-tap: unilateral **0.055 ± 0.027**, bilateral in-phase 0.051, antiphase 0.047 (p = 0.011). SCA-tap: 0.063 / 0.061 / 0.057 (p = 0.009). **Use as the lower anchor** — metronome-paced lab taps are the regularity ceiling, which is why your rhythmic band sits above these values.

[11] S. Boripuntakul, T. Kamnardsiri, S. R. Lord, S. Maiarin, P. Worakul, and S. Sungkarat, “Gait variability during abrupt slow and fast speed transitions in older adults with mild cognitive impairment,” *PLOS ONE*, vol. 17, no. 10, art. e0276658, 2022.
https://doi.org/10.1371/journal.pone.0276658

Supports: Supplies the **formal definition** you can cite for the statistic itself — CV = (SD / mean) × 100 applied to inter-event intervals of a repetitive human movement — and shows it discriminates a cognitively impaired group from controls.

Quotable figures: Step-time CV significantly higher in MCI vs control on the slow-to-fast transition, **p = 0.001**. Older adults with MCI vs healthy controls.

[12] N. A. Timmermans, R. Terranova, D. C. Soriano, H. Cagnan, Y. P. Raykov, I. G. Bucur, B. R. Bloem, R. C. Helmich, and L. J. W. Evers, "A generalizable and open-source algorithm for real-life monitoring of tremor in Parkinson's disease," *npj Parkinson's Disease*, vol. 11, art. 205, 2025.
https://doi.org/10.1038/s41531-025-01056-2

Supports: Large-scale validation of the general principle behind your rhythm test: detect a repetitive motor behaviour using features that **explicitly capture periodicity**, gated by a frequency-band threshold. Also a strong external-validation model for your Methodology.

Quotable figures: 24 + 24 training, **517 PD + 50 controls external validation**. Sensitivity 0.61 (SD 0.20), specificity 0.97 (SD 0.05); **peak-frequency gate 3–7 Hz**; test–retest ICC 0.98 for tremor time.

Repetitive motor behaviour from video — feasibility at clinical scale

[13] T. Barami, L. Manelis-Baram, H. Kaiser, M. Ilan, A. Slobodkin, O. Hadashi, D. Hadad, D. Waissengreen, T. Nitzan, I. Menashe *et al.*, "Automated Analysis of Stereotypical Movements in Videos of Children With Autism Spectrum Disorder," *JAMA Network Open*, vol. 7, no. 9, art. e2432851, 2024.
https://doi.org/10.1001/jamanetworkopen.2024.32851

Supports: Automated video detection and **counting** of repetitive stereotyped movements at clinical scale. Cite for "repetitive movements are reliably extractable from video trajectories" — not for the CV method, which this paper does not use.

Quotable figures: 220 children, 319 assessments, **580 hours of video**. Detection accuracy **92.53%** (95% CI 81.09–95.10). Strong correlation between algorithm-derived and manually documented movement counts and durations.

[14] S. S. Khan, P. K. Mishra, N. Javed, B. Ye, K. Newman, A. Mihailidis, and A. Iaboni, "Unsupervised Deep Learning to Detect Agitation From Videos in People With Dementia," *IEEE Access*, vol. 10, pp. 10349–10358, 2022.
https://doi.org/10.1109/ACCESS.2022.3143990

Supports: The most directly comparable deployment in the literature: **existing ceiling surveillance cameras in a long-term care dementia unit** used to flag clinically significant agitation. Same setting, same constraint, same class-imbalance problem as VisioSphere.

Quotable figures: Trained on ~24 h of normal activity, tested on 11 h containing annotated agitation events. **AUC(ROC) = 0.754.** Note the modest AUC — useful for calibrating panel expectations about what is achievable in this setting.

[15] P. K. Mishra, A. Iaboni, B. Ye, K. Newman, A. Mihailidis, and S. S. Khan, "Privacy-protecting behaviours of risk detection in people with dementia using videos," *BioMedical Engineering OnLine*, vol. 22, art. 4, 2023.
https://doi.org/10.1186/s12938-023-01065-3

Supports: Your ethics-chapter citation: anonymising the video to **skeleton keypoints** costs essentially nothing in detection performance versus raw RGB. This is the published justification for a pose-only pipeline in a care facility.

Quotable figures: Specialized Dementia Unit, Toronto Rehabilitation Institute; 15 wall-mounted cameras, 352×240 @ 30 fps; ~600 days of footage. RGB AUC(ROC) **0.822**; **skeleton 0.812**; segmentation mask 0.823. AUC(PR) 0.124–0.134 — honest evidence of how severe the imbalance is.

[16] A. Badawi, S. Elmoghazy, S. Choudhury, S. Elgazzar, K. Elgazzar, and A. M. Burhan, "Multimodal Detection of Agitation in People With Dementia in Clinical Settings: Observational Pilot Study," *JMIR Aging*, vol. 8, art. e68156, 2025.
https://doi.org/10.2196/68156

Supports: Current state of the art for camera-based agitation detection in a real inpatient dementia unit, with faces anonymised. Shows the video channel alone is viable and that fusing it with a wearable improves on either alone.

Quotable figures: 10 participants, single site. **Video: GRU 95% accuracy, AUC 0.99**; wearable Extra Trees up to 99% personalised. Agitation predicted ≥ 6 min in advance. **Present the AUC honestly — n = 10 pilot.**

[17] M. Al-Hammadi, H. Fleyeh, and I. Thomas, "Gait and movement analysis for discrimination between people with dementia and healthy control persons based on pose estimation and machine learning," *Journal of Alzheimer's Disease*, vol. 111, no. 1, pp. 222–239, 2026.
https://doi.org/10.1177/13872877261430001

Supports: Validates that **markerless 2D pose keypoints from ordinary RGB video** carry enough reliable motor signal in a dementia population to support person-level inference. Cite for pose-estimation validity in your population — it is not an agitation study.

Quotable figures: 64 people with dementia vs 67 controls, two memory clinics; 1920×1080 @ 30 fps; YOLOv8, **17 keypoints, mean keypoint visibility 0.77**. Best SVM **87% ± 5.1 accuracy**, 86.6% recall.

3. Validating the claim: "there is no single universal formula"

These eight are the evidence base for your defence argument — that direct calculation-level RRLs are scarce *because the field has no consensus formula*, and that the strongest recent work combines complementary measurements rather than perfecting one.

(a) The same model collapses on a different dataset

[18] V. Fula and P. Moreno, "Wrist-Based Fall Detection: Towards Generalization across Datasets," *Sensors*, vol. 24, no. 5, art. 1679, 2024.
https://doi.org/10.3390/s24051679

Supports: The cleanest single demonstration that a near-perfect published accuracy is a property of the dataset, not of the formula.

Quotable figures: Within-dataset AUC-ROC **99.25%**, recall 95.16%. Cross-dataset: UMAFall→UPFall AUC **68.34%**, recall **45.16%**; WEDAFall→UPFall AUC **65.23%**, recall 38.71%. Merging all three datasets restores recall to 90.57% — **only more diverse data recovers performance, not a better formula**.

(b) No consensus formula; reported accuracies are not comparable

[19] M. Ishaq, D. C. Guastella, G. Sutera, and G. Muscato, "A Systematic Review of Fall Detection and Prediction Technologies for Older Adults: An Analysis of Sensor Modalities and Computational Models," *Applied Sciences*, vol. 16, no. 4, art. 1929, 2026.
https://doi.org/10.3390/app16041929

Supports: The strongest single citation for your claim. The authors state a quantitative meta-analysis "was not feasible" because of "significant heterogeneity in sensor modalities, experimental protocols and performance metrics reported," and that "models that perform well on one dataset may not generalize to others."

Quotable figures: 130 studies. Reported accuracy spans **79.6–99.98%** (vision 79.6–99.98%, wearable 90.3–99.9%). **128 of 130 studies (98.5%) validated on simulated falls; only 2 tested real-world unanticipated falls.**

[20] P. Gorce and J. Jacquier-Bret, "Fall Detection in Elderly People: A Systematic Review of Ambient Assisted Living and Smart Home-Related Technology Performance," *Sensors*, vol. 25, no. 21, art. 6540, 2025.
https://doi.org/10.3390/s25216540

Supports: PRISMA systematic review, PROSPERO-registered, that carries **both halves of your claim at once**: measured heterogeneity, and measured superiority of multimodal approaches.

Quotable figures: 80 studies, **426 accuracy values pooled from 59 studies**. **63.5% used custom private datasets**, against 33 different open datasets. Wearable 91.1% / non-wearable 92.6% / **hybrid multimodal 94.6% accuracy, 97.6% sensitivity, 98.2% F1**; hybrid beat wearable-only at **$p < 0.05$**.

(c) Complementary measurements beat any single one

Reference **[5]** (Rehouma & Boukadoum 2025) is also your fusion ablation: combined FPR **3.6%** against **8.9%** vision-only and **11.3%** IMU-only — a 57–67% cut in false positives from adding a second complementary channel, with each component measured separately. Cite it in both sections.

(d) Real deployments perform nothing like lab benchmarks

[21] R. G. Curtis, C. Maher, K. A. Dankiw, B. Singh, and S. Bogomolova, "In-Home Evaluation of the Neo Care Artificial Intelligence Sound-Based Fall Detection System," *Future Internet*, vol. 16, no. 6, art. 197, 2024.
https://doi.org/10.3390/fi16060197

Supports: A **commercial single-modality** detector measured in real homes. This is the most useful number in your entire reference list: it is what happens when one measurement carries the whole decision. It also justifies your escalation ladder and circuit breaker.

Quotable figures: 18 households, mean age 75.3, 12-week trial. Sensitivity **75.00%** but specificity **6.80%**. **4,939 activations, of which 4,930 were false** (9 true) — roughly **one false alarm every 1.25 hours**. Variability attributed partly to the individual home's layout.

(e) Agitation specifically has no agreed measurement

[22] B. Wong, P. Wu, Z. Ismail, J. Watt, and Z. Goodarzi, "Detecting agitation and aggression in persons living with dementia: a systematic review of diagnostic accuracy," *BMC Geriatrics*, vol. 24, no. 1, art. 559, 2024.
https://doi.org/10.1186/s12877-024-05143-6

Supports: Even the **human** instruments disagree. If trained clinicians using validated scales cannot converge on what agitation is, a demand that your detector match a single universal formula is not a coherent standard. Quotable: "Until 2015, there lacked a consensus-based definition of agitation, and consequently a reference standard diagnosis."

Quotable figures: 36 studies. Specificity for the same construct ranges **28.6% to 98%** across instruments (NBRS 28.6–85.0%; PAS 57.1%; NPI-Spanish 97.8%; F-RAGE 98.0%). Meta-analysis impossible.

[23] M. Mbogori-Kairichi, A. Malioukis, R. S. Snead, J. Marczika, R. Ambalavanan, and G. Towett, "Digital biomarkers for early agitation detection in dementia: a scoping review of emerging wearable and smart technologies for personalized care," *Frontiers in Neurology*, vol. 17, art. 1683517, 2026.
https://doi.org/10.3389/fneur.2026.1683517

Supports: Shows the pattern in one table: high single-modality numbers are measuring narrow, easier problems; honest multimodal systems report far more sober figures. Quotable: “heterogeneity in how agitation was defined and measured across studies also constrains comparability.”

Quotable figures: 13 studies from 798 screened. Single-modality **91.69–98.6%** task-specific accuracy, versus multimodal **median AUC 0.87, sensitivity 71.32%, specificity 75.28%**. Multimodal systems were the more common design.

[24] N. Farina, L. Smith, M. Rajalingam, and S. Banerjee, “Implementation Outcomes for Agitation Detection Technologies in People with Dementia: A Systematic Review,” *Geriatrics*, vol. 10, no. 3, art. 70, 2025.
https://doi.org/10.3390/geriatrics10030070

Supports: The single best justification for your project existing. It does not validate detector performance — it validates the gap. Do not cite it for accuracy numbers; it reports none.

Quotable figures: 1,697 records screened, 19 included. **Median sample size 10.** 63% multimodal; **only 1 study (5%) was camera-based**; 26% in residential care. Key finding: “a general absence of primary data related to implementation outcomes.”

4. Carried forward — previously verified, still strong

These were verified in the earlier pass and remain valid for the updated system. Full details are in *VisioSphere_RRL_Abstracts.pdf*.

#	Source	Role
[25]	Y. Zhang *et al.*, “ByteTrack: Multi-Object Tracking by Associating Every Detection Box,” *ECCV*, 2022.	Your tracker. Associates low-confidence boxes — the mechanism that keeps a track alive through the occlusion your lying gate must survive.
[26]	A. Zeng *et al.*, “SmoothNet: A Plug-and-Play Network for Refining Human Poses in Videos,” *ECCV*, 2022.	Temporal filtering of keypoint trajectories — backs your 3-sample median filter before the rhythm test.
[27]	R. Beddiar, M. Oussalah, and B. Nini, “Fall detection using body geometry and human pose estimation in video sequences,” *JVCIR*, vol. 82, art. 103407, 2021.	Body-geometry ratios from pose — general support for the ratio family.
[28]	A. R. Inturi *et al.*, “A Novel Vision-Based Fall Detection Scheme Using Keypoints of Human Skeleton with Long Short-Term Memory Network,” *Sensors*, vol. 23, no. 14, art. 6283, 2023.	Keypoint-based fall detection with temporal modelling.
[29]	V. Dentamaro, D. Impedovo, and G. Pirlo, “Fall Detection by Human Pose Estimation and Kinematic Theory,” *ICPR*, pp. 2328–2335, 2021.	Angular kinematics for fall vs non-fall — backs the angular-velocity split (≥ 45 deg/s = FALL).
[30]	M. Nagaj *et al.*, *Smart Innovation, Systems and Technologies*, pp. 215–226, 2025.	Recent pose-based fall work.
[31]	J.-M. Gaya-Morey, C. Manresa-Yee, and J. M. Buades-Rubio, “Deep learning for computer vision based activity recognition and fall detection of the elderly: a systematic review,” *Applied Intelligence*, vol. 54, no. 19, pp. 8982–9007, 2024.	Systematic review of vision-based elderly fall detection — survey-level support for the heterogeneity argument.
[32]	A. C. Bayen *et al.*, “Reduction in Fall Rate in Dementia Managed Care Through Video Incident Review: Pilot Study,” *JMIR*, vol. 23, no. 6, art. e17551, 2021.	SafelyYou Guardian — the closest deployed comparator. 436 falls, 66 residents; ground time cut 29.6 min; falls > 1 h on floor fell 31% → 0. **Human-in-the-loop**, unlike VisioSphere.

Also carried and still valid for the alarm-fatigue argument: Kubitza *et al.*, *BMC Geriatrics* 22:582, 2022; Michels *et al.*, *BMC Nursing* 24:664, 2025; Blackburn *et al.*, *International Emergency Nursing* 62:101148, 2022; Gattani *et al.*, *Ageing Research Reviews* 113:102948, 2025; Raza *et al.*, *Engineering Applications of AI* 143:109809, 2025; Nabizade *et al.*, *Sensors* 26(16):5069, 2026; Chen *et al.*, *Symmetry* 12(5):744, 2020.

5. What the literature does NOT support — read before the defence

Three of your constants have **no citable 2020+ threshold behind them**. I searched specifically for each and found nothing that meets the bar. Claiming otherwise is the fastest way to lose the panel. The correct move is to present each as your own empirically tuned parameter, cite the *method* paper for the measurement, and put the tuning evidence in Methodology.

Constant	What is missing	What to say instead
LYING_HIP_EXTEND_MIN = 140°	No published hip-flexion degree cut-point separating supine/prone from seated in 2020+ pose-based posture classification. Everything found is either clinical ROM biomechanics or an ML classifier that consumes the angle without publishing a threshold.	Cite [3] (hip and knee angles *are* discriminative features) and [1] (angle thresholds of 30°/50° are publishable and effective). State 140° as your calibration.
LYING_STACK_MAX = 1.20 and the 2.5–3.0 / 1.4–2.0 / 1.1–1.8 / 0–1.0 bands	No paper publishes a torso-normalised vertical stack ratio with those bands. The bands are yours.	Cite [1] — its spine ratio has boundary 1.0 with an empirical split at 1.2, which is remarkably close to your 1.20 and is the nearest published analogue. Say so; it is a genuinely strong parallel.
AGITATION_OSC_MAX_CV = 0.55 / 0.40 and the CV 0.1–0.3 rhythmic band	No paper reports or justifies a numeric CV cut-off separating rhythmic from random human movement. Also: **no 2020+ peer-reviewed source states and measures CV ≈ 1 for a Poisson event train in a behavioural context** — that result is pre-2020 statistical theory.	Cite [10] and [11] for CV-of-intervals as the accepted rhythmicity statistic, and attribute the Poisson property to standard theory (Cox & Lewis, *The Statistical Analysis of Series of Events*, 1966) rather than to a recent paper. Your *mc_agitation.py* Monte Carlo table is the evidence for the cut-off — lead with it.

Citation correction required

"Kaur et al. 2023" in LYING_FIX_APPLY.md does not exist. Crossref (author = Kaur, 2022–2024, agitation / dementia / hand movement) and open-web searches return no such paper. The 30–90 rpm figures are from **Marshall et al. 2022 — reference [9]**. Two further corrections: the **6-second window is for movement *type***, while intensity (the rpm judgement) uses **12-second windows with 5-second overlap**; and Marshall used **20 healthy actors** simulating agitation, not people with dementia. Remove the fabricated citation from every document before the defence.

Rejected after review — considered and dropped

OmniFall (arXiv preprint — excluded despite being topically ideal); Tîrziu *et al.* and Iazzi *et al.* (no reported metrics); Ullah *et al.*, *Technologies* 13(5):184, 2025 (authors state "no formal numerical metrics were computed"); Elmoghazy *et al.*, *Alzheimer's & Dementia*, 2025 (conference abstract, not a full paper — cite [16] instead, same group); Koo *et al.*, *Sensors* 21(14):4638, 2021 (cross-dataset figures not extractable); Moutsis *et al.* and Jabbar *et al.* (accelerometer/wearable only); Iaboni *et al.*, *Alz. Dement. DADM* 14:e12305, 2022 (publisher returned 403 — topically ideal, **worth retrieving through your university library**).